

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Devices Using SMPS (Switched Mode Power Supply):

1. Desktop Computer (PC Tower)

Requires high efficiency and lightweight, multi-rail DC voltage regulation to power sensitive components (CPU, GPU, motherboard) without generating excess heat.

2. Laptop Power Adapter

Needs compact, highly efficient AC-to-DC conversion so the charger brick remains portable and doesn't overheat.

3. LED / Smart TV

Requires precise, stable DC supply voltages for thin internal components and backlights within a slim chassis.

Devices Using UPS (Uninterruptible Power Supply):

1. Wi-Fi Router / Modem

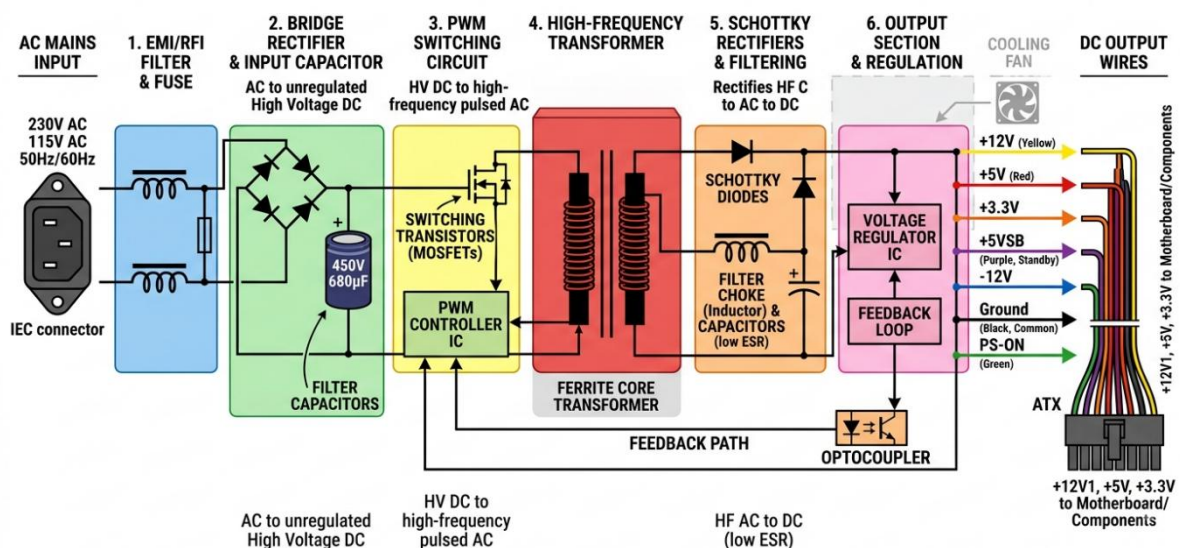
Needs instantaneous backup power during power outages to keep internet connections, active downloads, and smart home networks online without restarting.

2. Desktop Computer / College Server

Requires zero-delay power backup during outages to prevent immediate shutdown, hardware damage, and data loss before work can be saved.

2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.
 Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.

BLOCK DIAGRAM OF A TYPICAL DESKTOP PC SMPS UNIT



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Power Connector:



- Name: ATX 24-Pin Main Motherboard Power Connector
- Devices Powered: Primary power feed for the entire motherboard, chipset, RAM, onboard M.2 SSDs, PCIe slot power (up to 75W), and integrated peripherals.
- Safety Tip: Always press down and release the plastic retention latch fully before pulling, and pull directly on the plastic housing rather than yanking the wires, as pulling the cables can pull pins out of the connector housing.

2. SATA Power Connector:



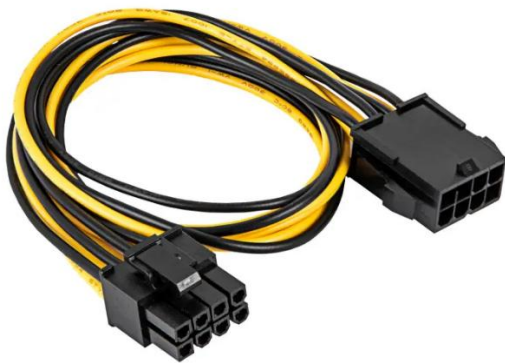
- Name: SATA (Serial ATA) Power Connector
- Devices Powered: 2.5-inch and 3.5-inch SATA Hard Drives (HDDs), SATA Solid State Drives (SSDs), Optical Disc Drives (CD/DVD/Blu-ray writer), and RGB/fan control hubs.
- Safety Tip: Insert and remove the connector completely straight to avoid snapping the thin, notch-shaped plastic guide on the SATA drive's power port.

3. Molex Connector:



- **Name:** 4-Pin Molex (Peripheral) Power Connector
- **Devices Powered:** Older IDE hard drives, optical drives, case fans, liquid cooling pump controllers, and legacy expansion accessories.
- **Safety Tip:** Ensure all four internal metal pins are properly aligned with the female sockets before pressing together; forcing misaligned pins can cause short circuits or wire dislodging.

4. PCIe 6/8-Pin (6+2 Pin) Connector:



- **Name:** PCIe 6-Pin / 8-Pin (6+2) Power Connector
- **Devices Powered:** Dedicated high-performance Graphics Processing Units (GPUs / graphics cards) requiring supplemental 12V power beyond the PCIe slot capacity.

- **Safety Tip:** Never substitute a CPU 8-pin (EPS) connector into a PCIe slot (or vice versa), as their voltage pinouts and plastic keying are inverted, which will cause a direct short circuit and ruin the GPU or motherboard.

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Pre-Installation & Safety

1. Ensure the power supply (SMPS) is turned off (**O** position on the rocker switch) and completely unplugged from the wall outlet.
2. Install the SMPS securely into its designated bay in the PC case using four standard 6-32 thumb screws.
3. Route all power cables through the cable management cutouts behind the motherboard tray to keep the main chamber clear.

Step 2: Motherboard Power Setup

1. Main System Power (ATX 24-Pin):

- **Cable/Connector: 24-Pin (or 20+4 Pin) ATX Power Cable**
- **Action:** Locate the large 24-pin socket on the right edge of the motherboard. Align the cable latch with the plastic notch on the motherboard connector, press down firmly until the latch clicks into place, and verify there is no visible gap between the plastic housings.

2. CPU Power (EPS 12V):

- **Cable/Connector: 8-Pin (or 4+4 Pin) EPS / CPU Power Cable**
- **Action:** Route the CPU power cable to the top-left corner of the motherboard. Match the square and rounded pin shapes, align the clip, and press firmly into the 4-pin or 8-pin CPU socket until it clicks.

Step 3: Graphics Card (GPU) Power Setup

- **Cable/Connector: PCIe 6+2 Pin Power Cable** (*or 12VHPWR / 16-Pin cable for newer NVIDIA GPUs*)
- **Action:**
 1. Combine the 6-pin and 2-pin clips into a unified 8-pin connector if your GPU requires 8-pin inputs.
 2. Plug individual PCIe power cables from separate PSU ports into each socket on the graphics card rather than daisy chaining a single split cable if the GPU consumes over 225W.
 3. Push the connectors completely in until the retention latches fully click to prevent loose contacts and thermal melting.

Step 4: Storage Devices Power Setup

1. **SATA HDDs / 2.5-inch SSDs:**
 - **Cable/Connector: SATA Power Cable** (ribbon cable with wide, flat "L"-shaped connectors)
 - **Action:** Align the "L"-shaped notch of the connector with the power port on the drive and press straight in without angling or twisting.
2. **M.2 NVMe SSDs (Note):**
 - **Cable/Connector:** *No power cable required.*
 - **Action:** M.2 drives draw power directly from the PCIe slot via the 24-pin motherboard connector connected in Step 2.

Step 5: Peripheral Devices & Final Check

1. **RGB/Fan Hubs (if applicable):**
 - Use either a **SATA Power Cable** or a **4-Pin Molex Cable** depending on the controller module's input requirements.

2. Post-Connection Review:

- Double-check that all modular cables are fully seated on **both ends** (the PSU side and the component side).
- Plug the AC power cord into the back of the SMPS, turn the switch to the **I** (ON) position, and test the system power button.